ORIGINAL ARTICLE

"Super-Spreaders" and Person-to-Person Transmission of Andes Virus in Argentina

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ABSTRACT

BACKGROUND

From November 2018 through February 2019, person-to-person transmission of Andes virus (ANDV) hantavirus pulmonary syndrome occurred in Chubut Province, Argentina, and resulted in 34 confirmed infections and 11 deaths. Understanding the genomic, epidemiologic, and clinical characteristics of person-to-person transmission of ANDV is crucial to designing effective interventions.

METHODS

Clinical and epidemiologic information was obtained by means of patient report and from public health centers. Serologic testing, contact-tracing, and next-generation sequencing were used to identify ANDV infection as the cause of this outbreak of hantavirus pulmonary syndrome and to reconstruct person-to-person transmission events.

RESULTS

After a single introduction of ANDV from a rodent reservoir into the human population, transmission was driven by 3 symptomatic persons who attended crowded social events. After 18 cases were confirmed, public health officials enforced isolation of persons with confirmed cases and self-quarantine of possible contacts; these measures most likely curtailed further spread. The median reproductive number (the number of secondary cases caused by an infected person during the infectious period) was 2.12 before the control measures were enforced and decreased to 0.96 after the measures were implemented. Full genome sequencing of the ANDV strain involved in this outbreak was performed with specimens from 27 patients and showed that the strain that was present (Epuyén/18–19) was similar to the causative strain (Epilink/96) in the first known person-to-person transmission of hantavirus pulmonary syndrome caused by ANDV, which occurred in El Bolsón, Argentina, in 1996. Clinical investigations involving patients with ANDV hantavirus pulmonary syndrome in this outbreak revealed that patients with a high viral load and liver injury were more likely than other patients to spread infection. Disease severity, genomic diversity, age, and time spent in the hospital had no clear association with secondary transmission.

CONCLUSIONS

Among patients with ANDV hantavirus pulmonary syndrome, high viral titers in combination with attendance at massive social gatherings or extensive contact among persons were associated with a higher likelihood of transmission. (Funded by the Ministerio de Salud y Desarrollo Social de la Nación Argentina and others.)

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Hantavirus pulmonary syndrome[1,2] is a rare, acute zoonotic disease with a wide distribution in the Americas.[3] The syndrome was described in 1993 in the United States[4] and can be caused by at least 24 distinct viruses[3] classified in the genus orthohantavirus (subfamily *Mammantavirinae*, family *Hantaviridae*, order *Bunyavirales*).[5] In Argentina and surrounding countries, most cases of Andes virus (ANDV) hantavirus pulmonary syndrome are caused by several members of the species *Andes orthohantavirus*: ANDV, Castelo dos Sonhos virus, Lechiguanas virus (LECHV), and Orán virus (officially classified strains)[5]; and Araucária virus, Araraquara virus, Bermejo virus, Buenos Aires virus, Juquitiba virus, Paranoá virus, and Plata virus (unclassified strains).[3,5] The incidence of death among persons with hantavirus pulmonary syndrome associated with ANDV is high and has ranged from 21 to 50% in Argentina.[6-8]

Humans generally become infected with orthohantaviruses through the inhalation of aerosolized secreta or excreta containing infectious virions originating from a rodent reservoir (e.g., the long-tailed colilargo [*Oligoryzomys longicaudatus*]). Before the discovery of ANDV in 1996, the route of orthohantavirus transmission was considered to be strictly zoonotic, resulting in dead-end human infections.[9] However, in 1996, an outbreak of hantavirus pulmonary syndrome caused by ANDV occurred in the small city of El Bolsón in Argentina and then expanded to other cities; the outbreak involved 16 epidemiologically linked cases. This outbreak became a focal point for orthohantavirus research because molecular and epidemiologic evidence suggested person-to-person transmission.[6,10,11] Although cases of hantavirus pulmonary syndrome caused by ANDV and its close relatives have been reported in Argentina and surrounding countries over the past two decades,[8,12-15] limited person-to-person transmission has been reported and has been observed mostly in southern Argentina and in Chile.[6-8,10,11,15-20] An important known risk factor for person-to-person transmission of ANDV is prolonged or close contact with symptomatic persons.[17,18,21]

With the outbreak of ANDV hantavirus pulmonary syndrome in Epuyén (Chubut Province, Argentina) in 2018 and 2019, ANDV received renewed attention with respect to possible person-to-person transmission. One patient was suspected to be the source of infection in several other patients, suggesting that an early super-spreading event initiated the subsequent transmission chains.[22] Super-spreading events are frequent in infectious-disease outbreaks and are the result of heterogeneities in the likelihood that a person will infect susceptible hosts and the varying degrees of transmission efficiency of the virus.[23,24] Factors related to characteristics of the host, the pathogen, and the setting of transmission all contribute to the occurrence of super-spreading events in outbreaks. Super-spreading events and sustained person-to-person transmission have not been reported previously in outbreaks of hantavirus pulmonary syndrome but have been observed in disease outbreaks caused by Ebola virus,[25] Middle East respiratory syndrome coronavirus, severe acute respiratory syndrome coronavirus (SARS-CoV), and SARS-CoV-2.[26,27] In the case of SARS-CoV, person-to-person transmission is most likely affected by host-related viral changes that occurred over time. We report the genetic, clinical, and epidemiologic features of the hantavirus pulmonary syndrome outbreak caused by the Epuyén/18–19 strain of ANDV in Epuyén, a small, rural village and municipality of approximately 2800 persons located in the Andean region of Chubut Province in southwestern Argentina.

METHODS

DATA COLLECTION AND ANALYSES

In addition to the cases of ANDV hantavirus pulmonary syndrome identified in Epuyén, epidemiologically linked cases were also reported in Río Negro Province, and one patient (Patient 21) identified in Chile had been infected by a patient from Epuyén.[28] Details regarding the case definition of ANDV hantavirus pulmonary syndrome are provided in Supplementary Appendix 1, available with the full text of this article at NEJM.org (unless otherwise noted, all supplementary figures and tables cited in the main article text are located in Supplementary Appendix 1). Written informed consent was obtained from all case patients or their proxy respondents. All procedures used in the investigation of the outbreak were approved by the ethics committee of the Instituto Nacional de Genética Médica at the Administración Nacional de Laboratorios e Institutos de Salud Dr. Carlos G. Malbrán (Buenos Aires).

Clinical and laboratory data were obtained at

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presentation for all patients identified in Argentina (33 patients) (Table 1). Supplementary Appendix 2 lists the maximum levels and the time that the highest level occurred for each laboratory measure. Antibody testing and reverse-transcriptase–quantitative polymerase-chain-reaction (RT-qPCR) assays were used to confirm ANDV infections in suspected cases. A case of laboratory-confirmed hantavirus pulmonary syndrome was defined as a suspected case in which the titers of IgM and IgG antibodies against ANDV were above the prespecified thresholds or as a case in which circulating ANDV RNA was detected by real-time RT-qPCR.[8,30] Cases were classified according to grades of disease severity, as previously described.[29] Data regarding epidemiologic factors and the close contacts of infected persons were retrospectively obtained by means of patient interviews at the clinic and at follow-up visits to patient homes by public health authorities throughout the study.

Details regarding sequencing methods, sequence reconstruction, phylogenetic analyses, haplotype networks, and viral population genomics are available in Supplementary Appendix 1. Serum and plasma cytokine profiles were established with a 48-Plex Bio-Plex Pro Human Cytokine Screening Panel (Bio-Rad Laboratories). Information regarding immunoassay methods and analyses is provided in Supplementary Appendix 1.

STATISTICAL ANALYSIS

Statistical analyses were performed with R packages "stats," "R0,"[31] "EpiEstim,"[32] "matrixstats," and "ggplot2." Univariable associations were compared with the use of the nonparametric Wilcoxon rank-sum test and were analyzed with Pearson's product-moment correlation test. Univariate logistic-regression models were also used to investigate factors associated with person-to-person transmission among patients with confirmed cases. Detailed information about statistical methods is provided in Supplementary Appendix 1.

To identify "super-spreaders," we estimated the Poisson distribution of the reproductive number (R [the number of secondary cases caused by an infected person during the infectious period, reported as the posterior median value]) of the outbreak with a stochastic range for the individual reproductive number (Z [the number of secondary infections caused by each patient, reported as an integer]), as previously described.[23,24] Only events of person-to-person transmission with a high certainty of infection at the time of the event were included in analyses and are reported as Z values. Patients were classified into different groups for several downstream statistical comparisons on the basis of calculated Z values.

RESULTS

PHYLOGENETIC, GENOMIC, AND IMMUNOLOGIC ANALYSES

Specimens were obtained from 28 of the 34 patients (82%) with confirmed cases related to the outbreak (Patients 1 through 20 and Patients 22 through 29); six specimens were obtained from patients with cases of hantavirus pulmonary syndrome unrelated to the outbreak in Epuyén who were from the provinces of Chubut, Neuquén, and Río Negro — regions in which ANDV is endemic. The Epilink/96 strain of ANDV was isolated from one specimen; this was the strain responsible for the 1996 outbreak in El Bolsón, approximately 40 km north of Epuyén.[10] RNA belonging to the S segment of the viral genome of ANDV was detected by RT-qPCR in all blood specimens (Table 1 and Supplementary Appendix 2).

Next-generation sequencing yielded complete or nearly complete ANDV genomic sequences (from the S, M, and L segments), with more than 98% coverage, from 27 of 28 available specimens (Table S3). Phylogenetic analysis confirmed that the Epuyén/18–19 strain belongs to the ANDV variant (Fig. 1A, Fig. S1, and Table S4). A median-joining haplotype network showed that the infections in 78% of patients (21 of 27 patients) shared a single viral haplotype that formed a central node from which six additional haplotypes spread out (Fig. 1C). Except for the haplotype of the specimen obtained from Patient 22, all the variant haplotypes were in terminal nodes (i.e., they were not transmitted to secondary cases). Sequences shared 99.8 to 100.0% identity with the other ANDV sequences involved in the outbreak (Table S5). No significant differences in subclonal single-nucleotide polymorphism frequencies or segment diversity were observed be-

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tween patients in different groups (super-spreaders vs. non–super-spreaders and all spreaders vs. nonspreaders) or between central and terminal haplotypes (P>0.05) (Fig. S2). Sequences identified in the outbreak shared approximately 94.1 to 100.0% identity with the ANDV RefSeq sequence (GenBank accession number, AF291702–4), with 27 nonsynonymous changes in total (Table S6).

The phylogenetic analysis confirmed that the outbreak was caused by a single spillover event (Fig. 1A). The Epuyén/18–19 sequences are most closely related to those recovered from an unrelated case of ANDV hantavirus pulmonary syndrome that occurred in 2018 (the NRC-6/18 strain) in a patient from a neighboring town and to the Epilink/96 strain.

EPIDEMIOLOGIC ANALYSIS

Three patients were the main drivers of the outbreak of ANDV hantavirus pulmonary syndrome in Epuyén and were the source of 64% of secondary cases (21 of 33 cases). The transmission and exposure events for all 33 secondary cases and four waves of human infection were reconstructed on the basis of the known and probable epidemiologic relationships among cases (Fig. 1B). Information regarding exposure types, dates, and the associated infected contacts is provided in Table 1 and Table S2.

Our findings traced the first person-to-person transmission event to a birthday party with approximately 100 guests. The index patient (Patient 1) attended the event for 90 minutes and was reportedly symptomatic at the time, with fever and malaise. Five persons (Patients 2 through 6) who had been seated close to Patient 1 reported onset of symptoms consistent with ANDV hantavirus pulmonary syndrome between 17 and 24 days after the party (Fig. S3). Patient 2 was the most likely source for six infections in other persons during the early prodromal phase because of his active social life. Patient 2 died 16 days after symptom onset, and his spouse (Patient 9) was febrile while attending his wake. An additional 10 persons who attended the wake and were in close contact with Patient 9 became ill between 14 and 40 days after the wake for Patient 2.

The remaining 12 patients were each in contact with at least one previously symptomatic patient. Four patients (Patients 10, 29, 33, and 34) may have been infected by more than one person. The mean (±SD) serial interval (the time from symptom onset in a primary case to symptom onset in an epidemiologically linked secondary case) was approximately 23±7 days (Fig. S4). Incubation periods ranged from 9 to 40 days.

On the basis of evidence from five reconstructed person-to-person transmission events, the route of infection in secondary cases was possibly through inhalation of droplets or aerosolized virions (Table 1 and Table S7). The time of symptom onset is also critical. In 17 of the 33 (52%) secondary transmission events, transmission from an infected person to a contact who later became infected could be accurately established as the day of onset of fever in the primary case, and the first day of fever was the only time during which there was an interaction between these patients (Table 1).

Although several high-risk medical procedures were performed in patients with ANDV hantavirus pulmonary syndrome, including orotracheal intubation and cleaning of bodily fluids such as vomit, diarrhea, and other secretions, no nosocomial infections were reported among health care workers who had been in direct or close contact with the patients at the health care facilities (Hospital Esquel Zonal and Epuyén Rural Hospital). Approximately 82 health care workers were exposed to symptomatic patients with confirmed ANDV infection at Hospital Esquel Zonal from December 2 to December 13, 2018. Of the 45 persons who worked in the intensive care unit and emergency department, only a small number used any form of personal protective equipment (including N95 respirators [N100 respirators for intubations and cleaning], goggles, and disposable laboratory coats) while they were in direct contact with an infected patient. Nonetheless, we identified one secondary nosocomial transmission event that occurred at Hospital Esquel Zonal, which is an advanced health care facility. Two additional nosocomial transmission events occurred at the smaller Epuyén Rural Hospital, for which information on the use of standard personal protective equipment was not available.

The posterior median R value across the entire outbreak was 1.19 (95% credible interval, 0.82 to 1.56), a finding that suggests that ANDV Epuyén/18–19 has sufficient transmission poten-

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Table 1. Characteristics of Cases of Andes Virus (ANDV) Hantavirus Pulmonary Syndrome.*

Characteristic	No./Total No. (%)
Sex of patient	
Male	14/34 (41)
Female	20/34 (59)
Age of patient	
≤17 yr	6/34 (18)
18–64 yr	22/34 (65)
≥65 yr	6/34 (18)
Disease outcome	
Patient survived	23/34 (68)
Patient died	11/34 (32)
Disease severity†	
Mild	15/34 (44)
Severe	19/34 (56)
Incubation period	
≤15 days	5/34 (15)
16–21 days	10/34 (29)
22–28 days	13/34 (38)
≥29 days	6/34 (18)
Type of exposure	
Zoonotic	1/34 (3)
Droplet or aerosol	6/34 (18)
Close contact	26/34 (76)
Unknown	1/34 (3)
No. of days patient had been symptomatic when infection was passed to secondary patient	
1 day	17/33 (52)
1–3 days	19/33 (58)
Unknown	14/33 (42)
Virologic results for ANDV infection	
Positive by RT-qPCR assay	33/33 (100)
Positive for IgM antibodies‡	33/33 (100)
Positive for IgG antibodies‡	31/33 (94)
Serologic findings§	
Leukocytosis	24/33 (73)
Neutrophilia	18/28 (64)
Lymphopenia	28/30 (93)
Thrombocytopenia	32/33 (97)
Elevated LDH level	33/33 (100)
Elevated AST level	30/32 (94)
Elevated ALT level	26/32 (81)
Irregular urea level	13/30 (43)

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Table 1. (Continued.)

Characteristic	No./Total No. (%)
Irregular creatinine level	17/30 (57)
Delayed prothrombin time	6/25 (24)
Decreased prothrombin index¶	5/25 (20)
Delayed aPTT	22/25 (88)

* Data are shown for 33 patients in Argentina and 1 patient in Chile. Percentages may not total 100 because of rounding. RT-qPCR denotes reverse-transcriptase quantitative polymerase chain reaction.
† Mild disease was defined as grade 1 or 2, and severe disease as grade 3 or 4.[29]
‡ The cutoff value for IgM was 0.3 (optical density at 450 nm) and for IgG, 0.55 (optical density at 450 nm).
§ The reference range used for leukocytes was 4500 to 11,500 per cubic millimeter; for neutrophils, 1300 to 6000 per cubic millimeter; for lymphocytes, 1500 to 3500 per cubic millimeter; for platelets, 150,000 to 450,000 per cubic millimeter; for lactate dehydrogenase (LDH), 313 to 618 U per liter; for aspartate aminotransferase (AST), 17 to 59 U per liter; for alanine aminotransferase (ALT), 21 to 72 U per liter; for urea, 15 to 45 mg per deciliter (5.4 to 16.1 mmol per liter); for creatinine, 0.66 to 1.25 mg per deciliter (58 to 110 μmol per liter) in women and 0.52 to 1.04 mg per deciliter (46 to 92 μmol per liter) in men[25]; for prothrombin time, 11.0 to 13.5 seconds; for prothrombin index, 60 to 130; and for activated partial thromboplastin time (aPTT), 26.0 to 37.5 seconds.
¶ The prothrombin index is the patient's prothrombin time expressed as a percentage of the normal value.

tial for self-sustaining transmission (R>1). The median weekly R value (6.56; 95% credible interval, 2.55 to 13.51) peaked from week 47 through week 48 (November 25 to December 1, 2018) and decreased to below 1.00 in late January 2019 (Fig. 1D). In comparison, the spread of the Epilink/96 strain of ANDV was estimated to have had a mean R value of 0.7 (90% confidence interval [CI], 0.2 to 1.1).[23]

Around December 31, 2018, after the 18th case of ANDV hantavirus pulmonary syndrome was confirmed, public health authorities ordered isolation measures for symptomatic patients and self-quarantine for high-risk case contacts (Supplementary Appendix 1). The median R value of secondary person-to-person transmission before the imposed control measures was 2.12 (95% credible interval, 1.24 to 3.35), and the median R value after control measures were implemented decreased to 0.96 (95% credible interval, 0.62 to 1.40).

The Z values were estimated for all patients involved in a secondary transmission event (Fig. 1B). A previously described formula was used to determine that the lower limit of Z for defining a super-spreader patient in our cohort was 4 (99th percentile), which characterizes Patients 1, 2, and 9 as super-spreaders.[23,24] To estimate factors in person-to-person transmission, patients were classified into four groups: super-spreaders (Z>4 [3 patients]), all spreaders (Z≠0 [10 patients]), non–super-spreaders (Z<4 [30 patients]), and nonspreaders (Z=0 [23 patients]) (Table S2).

CLINICAL CHARACTERISTICS

During the prodromal phase, patients with secondary cases of infection with the Epuyén/18–19 strain of ANDV presented with the typical clinical symptoms associated with ANDV hantavirus pulmonary syndrome (Table 2). The most common clinical feature after the prodromal phase was the rapid onset of acute respiratory distress, marking the onset of the cardiopulmonary phase, which was evidenced by different degrees of compromise of respiratory function (in 25 of 33 patients [76%]). Patients with severe cases had hypotension, progressive pulmonary edema, and hypoxia (often resulting in intubation), whereas patients with mild cases received only supplemental oxygen. Fatal cases were characterized by severe compromise in hemodynamic function, as evidenced by marked hypotension. Except for Patient 5, all person-to-person spreaders of ANDV presented with compromised respiratory function. The overall case fatality rate was 32% (11 of 34 patients), and the mean time from symptom onset to death was 6.7 days (interquartile range, 4 to 7). No significant differences or associations were found between disease severity and Z value (P>0.05) (Supplementary Appendix 3). Additional details regarding the clinical history of the patients and their treatments are provided in Supplementary Appendix 1.

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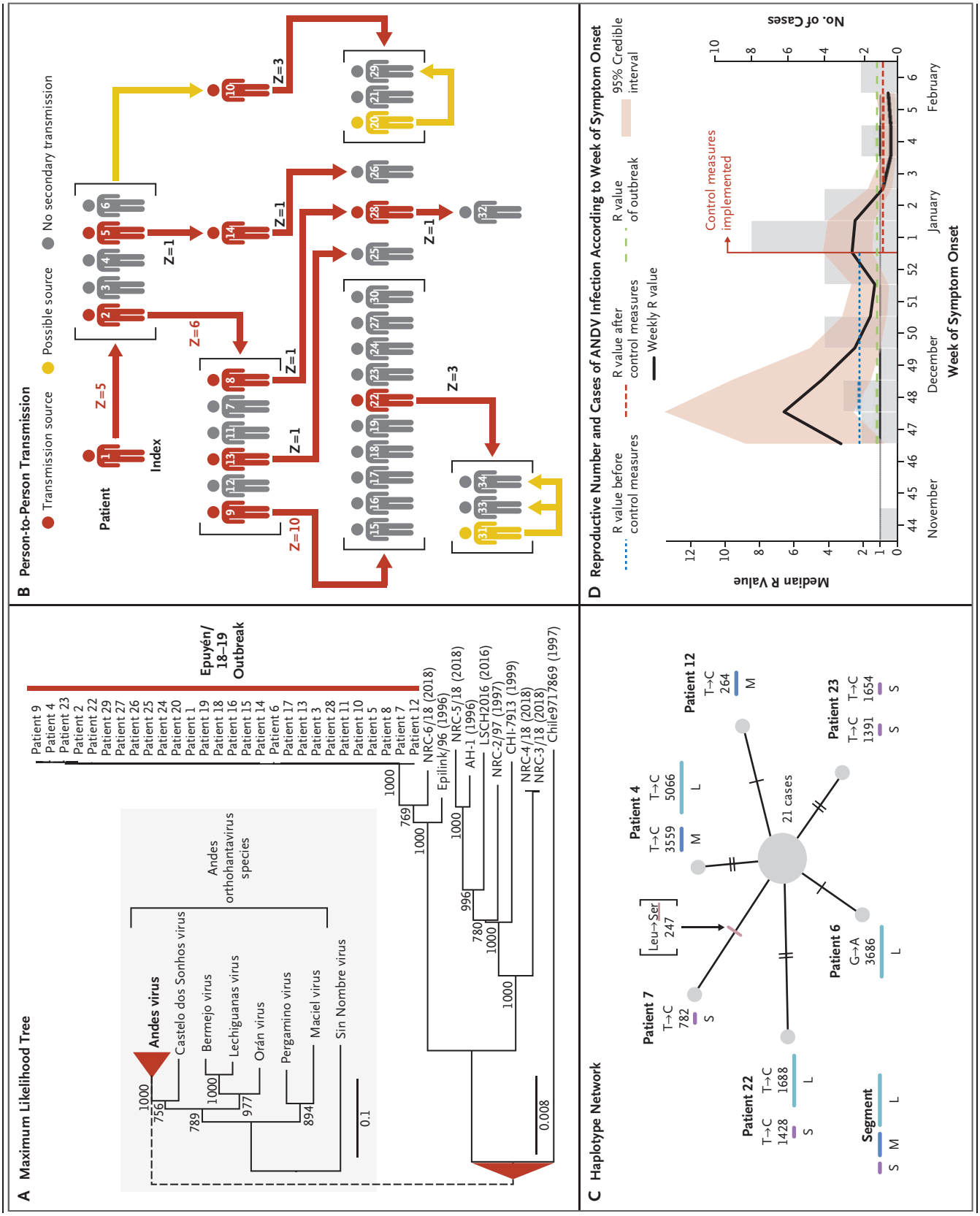

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Figure 1 (facing page). Epidemiologic Analysis of the 2018–2019 Outbreak of Andes Virus (ANDV) Hantavirus Pulmonary Syndrome and Transmission Dynamics in Humans.

Panel A shows the phylogenetic tree generated with the use of the maximum-likelihood method and estimated on the basis of S segment sequences (gray inset) and the concatenated S, M, and L segment sequences (ANDV subtree, right) with the use of 1000 bootstrap replications. The scale bars indicate the number of substitutions per site. Branch tips are labeled according to the strain designation and isolation date or patient number. Panel B shows an overview of ANDV transmission among patients. Patients who were the sources of transmission are indicated in red, and other possible but less likely transmission events between patients are indicated in yellow. Individual reproductive numbers (Z [the number of secondary infections caused by each patient, reported as an integer]) are shown where transmission events are indicated. Patients depicted in gray were not associated with evidence of extended transmission chains. Panel C shows a median-joining haplotype network generated with the use of concatenated segment sequences from 27 ANDV specimens. The loci of the nucleotide (black hash marks) and the amino acid (pink lines) are shown for each terminal node, which represents a single case. The S, M, and L segments of ANDV are represented by purple (S segment), dark blue (M segment), and light blue (L segment) bars. Panel D shows the reproductive number (R [the number of secondary cases per patient with a confirmed case]) and the incidence (gray shading) of ANDV hantavirus pulmonary syndrome of 34 cases over a 15-week period. The median R value was estimated for each week (solid black line) with 95% credible intervals (pink shading). The blue dashed line indicates the median R value before control measures were implemented (R=2.12) by the Ministry of Health, and the red dashed line indicates the R value after control measures were implemented (R=0.96); these values are shown for each 1-week period, where possible. The median R value for the entire 15-week period was 1.19. The thin black line indicates where R=1.00.

Table 2. Clinical Findings in Patients with ANDV Infection.

Clinical Finding	No./Total No. (%)
Prodromal symptoms	
Fever	33/33 (100)
Headache	19/33 (58)
Myalgia	19/33 (58)
Asthenia	22/33 (67)
Gastrointestinal manifestations	18/33 (55)
Abdominal pain	17/33 (52)
Nausea or emesis	10/33 (30)
Diarrhea	6/33 (18)
Odynophagia	3/33 (9)
Blurred vision	2/33 (6)
Cardiopulmonary outcomes	
Compromised hepatic function	32/33 (97)
Compromised respiratory function	25/33 (76)
Compromised renal function	18/30 (60)
Shock	11/33 (33)

LABORATORY STUDIES

Patients had elevated lactate dehydrogenase (LDH), aspartate aminotransferase (AST), alanine aminotransferase (ALT), urea, and creatinine levels; prolonged activated partial thromboplastin time; elevated prothrombin index (the patient's prothrombin time expressed as a percentage of the normal value); and prolonged prothrombin time (Table 1 and Supplementary Appendix 3). Hematologic abnormalities, including thrombocytopenia, lymphopenia, leukocytosis, and neutrophilia, were also observed in the majority of cases. Z values were positively correlated with ALT levels (r=0.51; 95% CI, 0.20 to 0.73) and longer prothrombin response times (r=0.52; 95% CI, 0.16 to 0.72) and were negatively correlated with the prothrombin index (r=−0.41; 95% CI, −0.69 to −0.02). Patients with higher ALT levels and more severe thrombocytopenia were also more likely to be super-spreaders (odds ratio for high ALT level, 1.6; 95% CI, 1.1 to 2.7; odds ratio for low platelet count, 2.9; 95% CI, 1.2 to 13.2).

Among the first 18 patients with confirmed infection (who had not yet been isolated), those with higher viral loads were more likely to spread infection to another person (odds ratio for elevated viral load, 1.7; 95% CI, 1.2 to 3.6). The viral load in patients before isolation was also positively correlated with patient age (r=0.46; 95% CI, 0.14 to 0.70). The median age of the patients was 38 years (interquartile range, 27 to 58). In addition, positive correlations were found between Z values and LDH (r=0.54; 95% CI, 0.09 to 0.80), AST (r=0.56; 95% CI, 0.10 to 0.82), and ALT levels (r=0.63; 95% CI, 0.21 to 0.85) in patients before they were isolated. The difference in laboratory measures before and after isolation could be affected by the fact that the first 18 patients were treated at health care facilities later after the onset of their initial symptoms (median, 4 days; 95% CI, 2.8 to 4.6) than the other patients (median, 2 days; 95% CI,

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Table 3. Serum Cytokine Profiles of Patients with ANDV Infection.*

Biomarker	All Spreaders (N=18)	Nonspreaders (N=33)	Odds Ratio (95% CI)	Super-Spreaders (N=5)	Non–Super-Spreaders (N=46)	Odds Ratio (95% CI)
	pg/ml			*pg/ml*		
Cytokines						
IL-1β	46.54±97.32	15.15±7.79	2.43 (1.28–5.48)	20.26±2.63	26.88±62.22	0.97 (∞–1.11)
IL-18	456.90±202.45	362.57±215.66	2.82 (0.73–12.20)	473.84±264.22	387.38±209.54	2.33 (0.28–16.00)
IL-13	15.40±9.80	11.90±NC	1.61 (0.61–4.68)	10.34±4.26	13.55±NC	0.55 (0.03–2.68)
IL-10	113.29±88.18	118.00±NC	0.96 (0.65–1.39)	62.29±41.70	122.41±NC	0.44 (0.12–1.02)
Chemokines						
MIP-1β	557.23±1116.10	304.89±60.20	1.09 (0.97–∞)	250.34±20.82	409.56±697.90	0.09 (0.00–0.81)
TRAIL	128.38±96.95	118.93±40.63	1.12 (0.70–1.79)	76.32±28.60	127.26±66.44	0.19 (0.02–0.80)
Growth factors						
PDGF-BB	3005.54±2061.08	3405.22±2890.86	0.97 (0.83–1.08)	1566.92±1022.70	3448.64±2672.30	0.53 (0.26–0.88)
GRO-α	593.66±123.33	671.57±238.06	0.76 (0.46–1.08)	528.53±65.93	656.63±213.51	0.38 (0.08–1.00)
SCF	233.92±66.24	203.10±72.44	1.20 (0.94–1.58)	245.64±96.40	210.53±68.45	1.21 (0.83–1.73)
SCGF-β	206789.86 ±66292.25	256201.74 ±60504.82	0.27 (0.09–0.73)	169924.61 ±38430.51	246244.61 ±64655.76	0.10 (0.01–0.59)

* Plus–minus values are means ±SD. Data are shown for specimens obtained on any day. Super-spreaders were considered to be patients whose Z value (the number of secondary infections caused by each patient, reported as an integer) was greater than 4. NC indicates that the value could not be calculated because of insufficient data. Odds ratios and 95% confidence intervals were estimated for the association between person-to-person transmission and elevated levels of biomarkers. P values were calculated with the use of the rank-sum test for the comparisons between variables. GRO-α denotes growth regulated oncogene α, IL interleukin, IL-1β interleukin-1β, MIP-1β macrophage inflammatory protein 1β, PDGF-BB platelet-derived growth factor BB, SCF stem cell factor, SCGF-β stem cell growth factor β, and TRAIL tumor necrosis factor–related apoptosis-inducing ligand.

1.8 to 3.1). Differences may also be attributed to very different clinical findings in the 3 patients who were super-spreaders than in non–super-spreaders (Supplementary Appendix 1).

SEROLOGIC ANALYSIS

Specific immunologic profiles may indicate whether single or multiple cytokines could help to identify patients with a higher likelihood of extending the chain of person-to-person transmission (Table 3 and Figs. S5 and S6 and Supplementary Appendix 4). Bivariate analyses revealed differences in the levels of interleukin-1β, interleukin-18, interleukin-13, stem cell factor, and stem cell growth factor β between all spreaders and nonspreaders (Table 3). Patients with higher levels of interleukin-1β (odds ratio, 2.4; 95% CI, 1.3 to 5.5) or a less pronounced elevation of stem cell growth factor β (odds ratio, 0.27; 95% CI, 0.09 to 0.73) were more likely to be spreaders than nonspreaders. Notably, stem cell growth factor β and the Z value in patients with the Epuyén/18–19 strain of ANDV were negatively correlated (r=−0.41; 95% CI, −0.61 to −0.15). Additional results are provided in Supplementary Appendix 1 and Supplementary Appendix 5.

DISCUSSION

The outbreak of ANDV hantavirus pulmonary syndrome in Epuyén in 2018 and 2019 shows evidence of person-to-person transmission of an orthohantavirus. The outbreak involved up to four generations of infections in humans and resulted in more cases than previous person-to-person outbreaks of ANDV.[16] Previous studies have used partial ANDV genomic information and contact tracing to reconstruct person-to-person transmission chains.[10,17] We used next-generation sequencing technologies to analyze

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ANDV transmission patterns on the basis of full genomic information from most of the cases in this outbreak (82%), which produced a more complete transmission network. On the basis of both the epidemiologic and genomic investigations of person-to-person transmission events, it appears that inhalation of droplets or aerosolized virions may have been the routes of infection.

The estimated difference in the R value after the public health authorities in Argentina established enforced self-quarantine of case contacts and the resulting reduction in the number of cases suggest that these measures limited the person-to-person spread of ANDV and therefore may have prevented additional super-spreading events. In addition, these measures reduced the number of potential case contacts, which may have allowed high-risk household contacts to be attended to and treated and may have resulted in detection of infection in the earlier stages of progression of hantavirus pulmonary syndrome.[18] However, this also limited our ability to rule out alternative avenues of transmission in multipatient homes.

Laboratory findings were generally consistent with previous reports.[8,33-36] The Z value was correlated with increasing ALT levels in the blood, longer prothrombin response times, and decreasing prothrombin index. These correlations suggest that person-to-person spread was related to a high viral load and more compromised liver function in the infected patient. Several differences between the patterns of immune response in all spreaders and nonspreaders were also notable, but the substantial up-regulation of the cytokines interleukin-18 (proinflammatory), interleukin-1β (inflammatory response), interleukin-13 (Th2), stem cell factor (hematopoiesis), and stem cell growth factor β (hematopoiesis) show a broad but dysregulated immune response. We did not find an association between the observed dysregulation and a particular disease outcome, specific immunologic pathway, or potential for secondary transmission.

Overall, these observations indicate that regardless of disease severity or the immune response of human hosts, patients with ANDV hantavirus pulmonary syndrome should be isolated and their cases carefully managed. Frontline care workers should use appropriate personal protective equipment to help prevent nosocomial transmission. Previous ANDV hantavirus pulmonary syndrome outbreaks in 1996[6] and 2011[37] included person-to-person nosocomial transmission involving health care workers, whereas outbreaks in 1999[38] and 2002[17] did not. However, our data suggest that three events of nosocomial transmission, although not involving health care personnel, occurred inside hospital facilities during the outbreak in Epuyén in 2018 and 2019.

The super-spreading capability of the ANDV Epuyén/18–19 strain shows a facility (R>2) for sustaining continuous chains of transmission if no control measures are enforced.[39] The person-to-person transmission calls for a careful evaluation of the epidemic potential and a biologic risk assessment of ANDV strains and viruses in the orthohantavirus genus. The absence of evidence for ANDV adaptation within or between hosts or for differences in viral diversity between spreaders and nonspreaders indicates that permissive ecology and social factors have a more substantial influence than genetic changes in sustaining person-to-person transmission in human hosts. Although the Epuyén/18–19 and Epilink/96 sequences were sampled 22 years apart, few genomic mutations were identified among the strains involved in the outbreak that were transmitted from person to person. In addition to their capabilities for sustaining person-to-person transmission, the genomic similarity of the Epuyén/18–19 and Epilink/96 strains suggests that they share the baseline genetic traits for successful cross-species transmission (facilitating spillover into humans) and transmission within a species (facilitating person-to-person transmission) without a need for adaptation to improve viral fitness, while a large number of contacts per infected patient and a sufficient critical community size (i.e., social gatherings and extensive contact) fueled early super-spreading events.[40-43] Genetic factors that facilitated person-to-person transmission may have been maintained in reservoir rodent host populations before the cross-species transmission that occurred in October 2018.

These data reinforce the need to study the patterns of ANDV emergence, transmission, and pathology to improve public health response strategies and to provide the most appropriate patient care in areas in which the virus is endemic.

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APPENDIX

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